

JEE-Main-02-04-2026 (Memory Based)

[EVENING SHIFT]

Physics

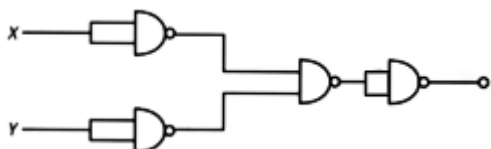
Question: In equilateral prism the path of a ray is shown. Determine μ

Options:

- (a) 1.71
- (b) 1.52
- (c) 1.39
- (d) 1.84

Answer: (c)

Question: What is the equivalent gate for the circuit?



Options:

- (a) AND gate
- (b) OR gate
- (c) NAND gate
- (d) NOR gate

Answer: (d)

Question: A soap bubble of radius $r = 1$ mm, liquid of density $\rho_l = 2000 \text{ kg/m}^3$. At the instant, the bubble is rising upward with constant velocity $v = \frac{1}{2} \text{ cm/s}$. Find coefficient of viscosity (η).

Options:

- (a) $\frac{2}{9} N - S/m^2$
- (b) $\frac{4}{9} N - S/m^2$
- (c) $\frac{2}{3} N - S/m^2$
- (d) $\frac{8}{9} N - S/m^2$

Answer: (b)

Question: Dimensions of G (Universal gravitational constant) in terms of h (Plank's constant, length) will be

Options:

- (a) $h^{-1}Lm^{-2}t$
- (b) $hL^{-1}m^2t$
- (c) $hLm^{-2}t^{-1}$
- (d) $h^{-1}L^{-1}m^2t^{-1}$

Answer: (c)

Question: Position of a particle is given by $x = a\sin(50t + \pi/3)$. If speed and acceleration is zero, for the first time at time t_1 & t_2 respectively. Then t_1 & t_2 are

Options:

- (a) $\frac{\pi}{150}, \frac{\pi}{300}$
- (b) $\frac{300}{\pi}, \frac{75}{\pi}$
- (c) $\frac{50}{\pi}, \frac{150}{\pi}$
- (d) $\frac{300}{\pi}, \frac{150}{\pi}$

Answer: (b)

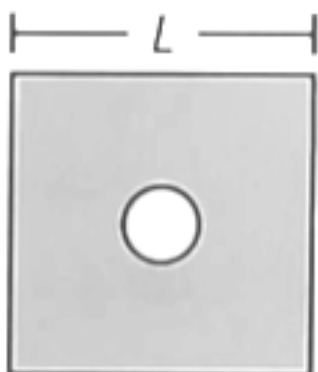
Question: In Bohr's atomic model. Find ratio of magnetic field produced at center by electron in 2nd orbit and 4th orbit.

Options:

- (a) 8
- (b) 4
- (c) 32
- (d) 16

Answer: (c)

Question: A small circular loop of radius r is completely enclosed within a large square loop of length L . Both of the loops are concentric and co-planar. (Also $L \gg r$). Find coefficient of mutual induction.



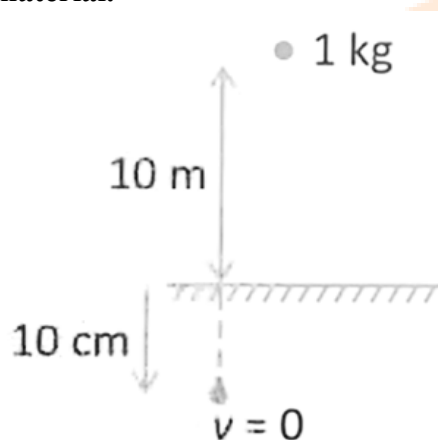
Options:

- (a) $M = \frac{\mu_0 r^2 2\sqrt{2}}{L}$

- (b) $M = \frac{\mu_0 L^2}{\sqrt{2}r}$
- (c) $M = \frac{2\mu_0 r^2}{L}$
- (d) $M = \frac{\mu_0 r^2}{2L}$

Answer: (a)

Question: A ball is released from rest as shown in figure. Ball comes to rest after moving 10 cm below the surface. Find the average force applied by the floor material.

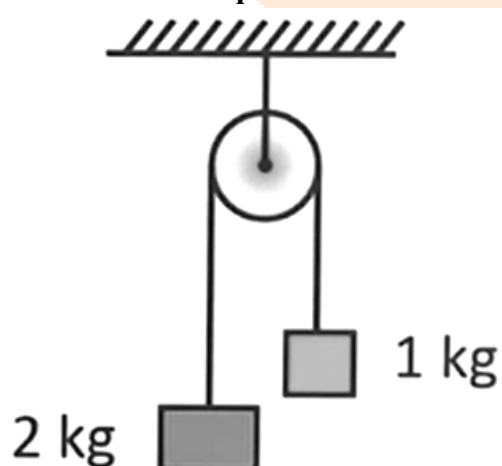


Options:

- (a) 800 N
(b) 1000 N
(c) 500 N
(d) 1200 N

Answer: (b)

Question: Find displacement of center of mass after $t = 2$ sec.

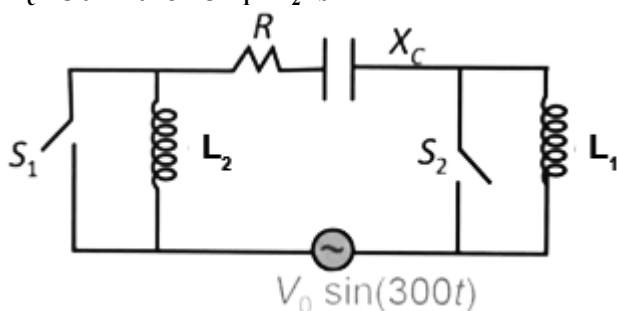


Options:

- (a) 20/9
- (b) 10/9
- (c) 25/9
- (d) 5/9

Answer: (a)

Question: In a given AC circuit if only switch S_1 is closed then phase difference is 30° while if only switch S_2 is closed then phase difference is 60° . Current lags in both cases. If $X_c = 30 \Omega$ then $3L_1 - L_2$ is

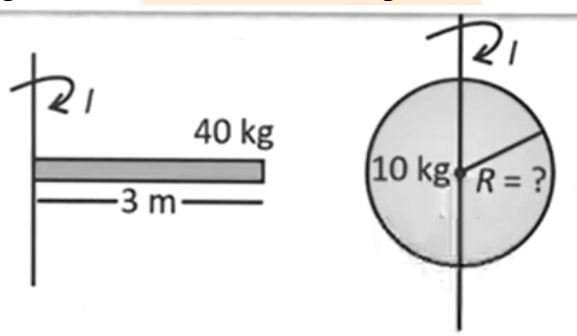


Options:

- (a) 0.3 H
- (b) 0.5 H
- (c) 0.2 H
- (d) 10 H

Answer: (c)

Question: A rod of length 3 m and mass 40 kg has same moment of inertia as that of solid sphere. Find radius of solid sphere.

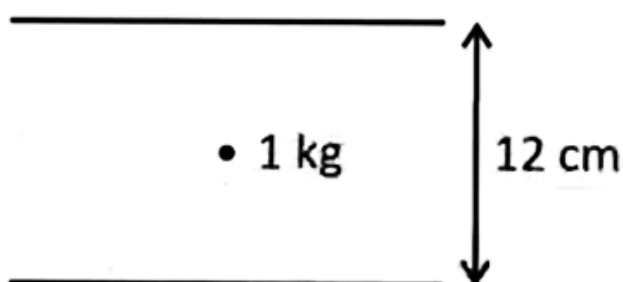


Options:

- (a) 5/2
- (b) $\sqrt{30}$
- (c) 3/2
- (d) $\sqrt{5}$

Answer: (b)

Question: A ball of mass 1 kg and charge 9 nC is in equilibrium between two parallel plates. Find potential difference between the plates.



Options:

- (a) $24 \times 10^3 \text{ V}$
- (b) $6 \times 10^6 \text{ V}$
- (c) $\frac{4}{3} \times 10^8 \text{ V}$
- (d) $12 \times 10^5 \text{ V}$

Answer: (b)

Question: A point mass just starts approaching earth from a large distance. Speed of impact of point of mass-to planet's surface is _____ .

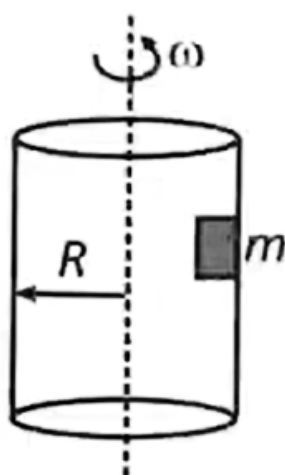
[Planet is 9 times heavier and with radius 4 times compared to earth]

Options:

- (a) 25 km/s
- (b) 10 km/s
- (c) 17 km/s
- (d) 5 km/s

Answer: (c)

Question: In the situation shown a small mass 'm' is rotating along with the cylinder with angular speed $\omega = 5 \text{ rad/s}$. The radius of the cylinder is $R = \frac{1}{2} \text{ m}$. Find the minimum value of friction coefficient between inner surface of cylinder and the mass 'm' so that it does not slip.



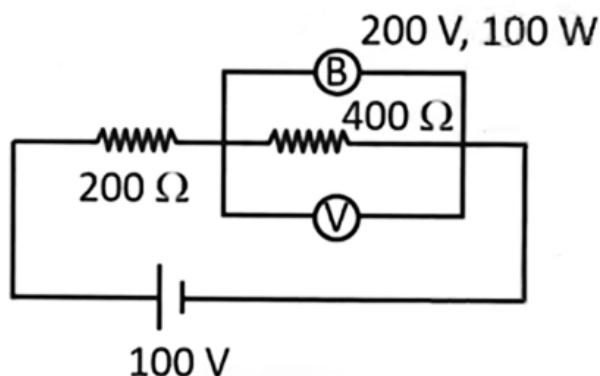
Options:

- (a) 5/4
- (b) 5/2
- (c) 2/5

(d) $\frac{4}{5}$

Answer: (d)

Question: In a DC circuit a bulb (200 V, 100 W) is connected with 2 resistances and an ideal voltmeter as shown. Reading of voltmeter is V volt then V is



Options:

- (a) 50 V
- (b) 25 V
- (c) 30 V
- (d) 40 V

Answer: (a)

Question: Find the work done when radius of a soap bubble is increased from 2 cm to 6 cm.

Surface tension of soap 0.03 N/m.]

Options:

- (a) 6 mJ
- (b) 2.4 mJ
- (c) 3.6 mJ
- (d) 2.1 mJ

Answer: (b)

Question: Two point charges $q_1 = +8\ \mu\text{C}$ and $q_2 = -2\ \mu\text{C}$ are placed at $x = 2$ and $x = 4$ respectively. Consider two spherical Gaussian surfaces S_1 and S_2 centered at the origin of radius 3 m and 5 m respectively. Electric flux through S_1 and S_2 is ϕ_1 and ϕ_2 then, $\frac{3\phi_1}{\phi_2}$ _____

Options:

- (a) 1
- (b) 4
- (c) 0
- (d) 2

Answer: (b)

Question: Temperature of a gas is increased from 10 k to 20 k isochorically. Find heat absorbed by the 5 moles of the gas ($C_p = 8\ \text{cal/mol k}$)

Options:

- (a) 340 J

- (b) 120 J
- (c) 520 J
- (d) 1260 J

Answer: (d)

Question: The intensity of sunlight, collected from equiconvex lens of $R = 60$ cm, is observed to be maximum at distance of 30 cm from lens, then refractive index of material of lens is

Options:

- (a) 1
- (b) 3
- (c) 2
- (d) 5

Answer: (c)

Question: A mixture of 13.2 gram of CO_2 & O_2 gas is maintained at pressure = 100 kpa, volume $v = 8310 \text{ cm}^3$ and temperature $T = 300 \text{ K}$. Find number of moles of CO_2 & O_2 respectively

Options:

- (a) $2/9$, $1/9$
- (b) $1.9/9$, $1.1/9$
- (c) $1.6/9$, $1.4/9$
- (d) $1.2/9$, $1.8/9$

Answer: (b)

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[EVENING SHIFT]
Chemistry

Question: Arrange following complexes in increasing order of CFSE (Δ_0)

- (a) $[\text{Co}(\text{H}_2\text{O})_6]^{2+}$
 (b) $[\text{Co}(\text{H}_2\text{O})_6]^{3+}$
 (c) $[\text{Co}(\text{en})_3]^{3+}$

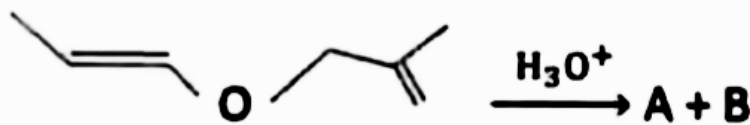
Options:

- (a) $c > a > b$
 (b) $c > b > a$
 (c) $a > b > c$
 (d) $b > a > c$

Answer: (b)

Solution: (b) $c > b > a$

Question: Consider the reaction given below:



A gives positive Fehling's test. Choose the correct relation.

Options:

- (a) Molar weight of A and molar weight of b are same
 (b) Molar weight of A is greater than molar weight of B
 (c) Molar weight of B is greater than molar weight of A
 (d) None of these

Answer: (c)

Solution: (c) Molar weight of B is greater than molar weight of A

Question: 20 mL of 0.2 M HA (weak monoprotic acid) is titrated with 10 mL of 0.2 M NaOH solution. pH of solution at 25°C is, (pK_a of weak acid is 4.76)

Options:

- (a) 9.24
 (b) 5.24
 (c) 4.76
 (d) 9.76

Answer: (c)

Solution: (c) 4.76

Question: Molarity of H_2SO_4 solution is 4.9 M. If density of solution is 1.40 g/ml, then molarity and mole fraction of solute in solution is

Options:

- (a) $m = 5.34$, $\mathcal{X}_{\text{solute}} = 0.088$

(b) $m = 5.34$, $\mathcal{X}_{\text{solute}} = 0.072$

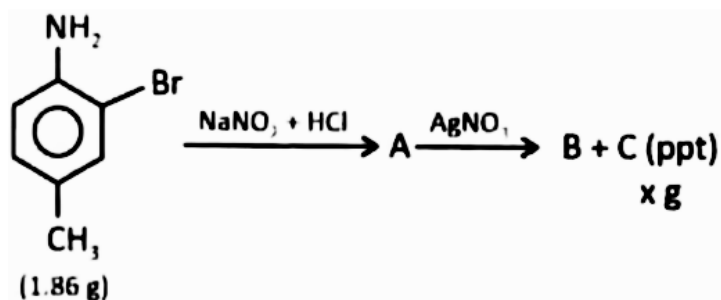
(c) $m = 5.21$, $\mathcal{X}_{\text{solute}} = 0.088$

(d) $m = 5.21$, $\mathcal{X}_{\text{solute}} = 0.072$

Answer: (a)

Solution: (a) $m = 5.34$, $\mathcal{X}_{\text{solute}} = 0.088$

Question:



‘C’ is a white curdy ppt. Find 10(x) (Nearest integer)

Options:

Answer: (14)

Question: SF_4 is isostructural with

(I) XeO_2F_2

(II) CH_4

(III) IF_4^+

(IV) BrF_4^-

Options:

(a) (I), (III), (IV) only

(b) (I), (III) Only

(c) (II), (IV) Only

(d) (I), (II), (III), (IV)

Answer: (b)

Solution: (b) (I), (III) Only

Question: The correct set that contains all kinds (basic, acidic, amphoteric and neutral) in Oxides is

Options:

(a) Na_2O , N_2O , Al_2O_3 and CO

(b) Al_2O_3 , As_2O_3 , CO and NO

(c) K_2O , Cl_2O_7 , As_2O_3 and NO

(d) Na_2O , K_2O , Al_2O_3 and As_2O_3

Answer: (c)

Solution: (c) K_2O , Cl_2O_7 , As_2O_3 and NO

Question: Among Fe^{2+} , Fe^{3+} , Cr^{2+} and Zn^{2+} , the ion that shows positive Borax bead test and with highest ionisation enthalpy is:

Options:

(a) Zn^{2+}

(b) Fe^{2+}

(c) Fe^{3+}

(d) Cr^{2+}

Answer: (c)

Solution: (c) Fe^{3+}

Question: The complete combustion of x gram of an Organic compound gave 0.25 g of CO_2 and 0.12 g of H_2O . If the % of carbon is 25% and of hydrogen is 4.89% then $x =$ $\text{---} \times 10^{-3}$ g (Nearest integer) (molar mass of C: H and O are 12, 1 and 16 g mol^{-1} respectively,

Options:

(a) 273

(b) 27.30

(c) 227

(d) 27

Answer: (a)

Solution: (a) 273

Question: For the reaction $2\text{A} \rightarrow 4\text{B} + \text{C}$. At 30 minutes the total pressure is 300 mm of Hg and after infinite time the total pressure is 600 mm of Hg. Then the pressure of C at $t = 30$ minutes will be

Options:

(a) 20 mm Hg

(b) 40 mm Hg

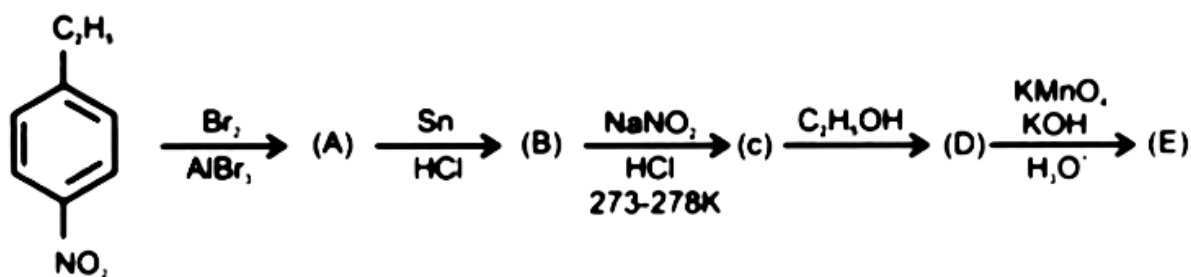
(c) 60 mm Hg

(d) 10 mm Hg

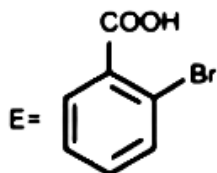
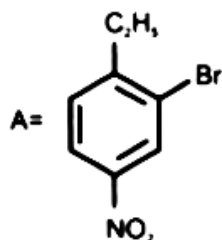
Answer: (a)

Solution: (a) 20 mm Hg

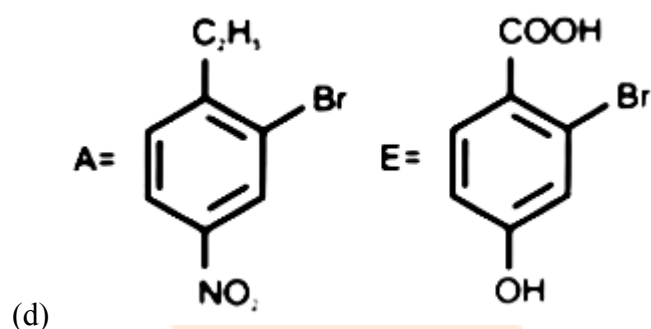
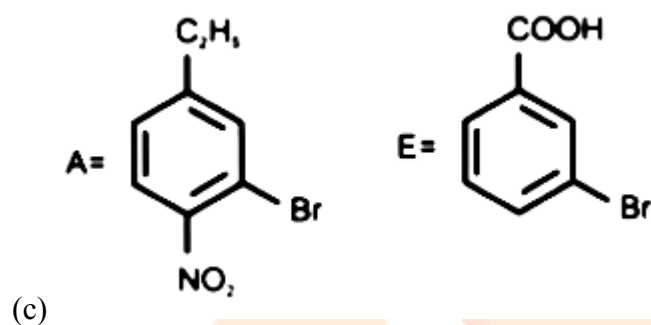
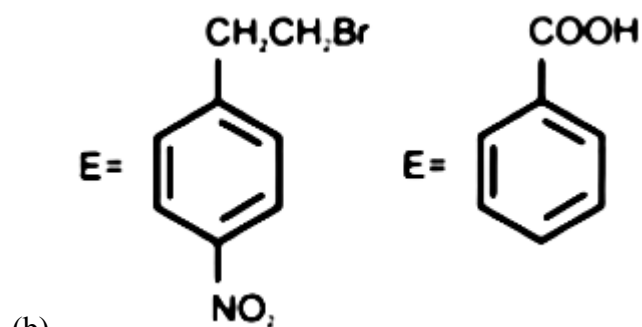
Question:



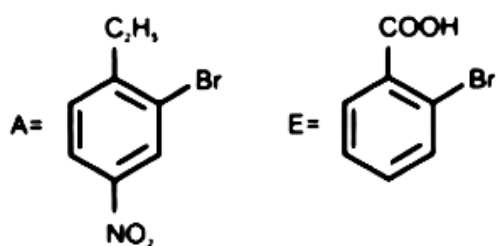
Options:



(a)



Answer: (a)
 Solution: (a)



Question: Two solutions of protein ($M_w = 50,000$ g) are prepared separately

Solution A: 1 g protein in 0.5 L solution

Solution B: 2 g protein in 1.0 L solution

When the two solutions are mixed at 300 K. Find total osmotic pressure:

($= 0.08 \text{ Lit-atm K}^{-1} \text{ mol}^{-1}$)

Options:

(a) 9.8×10^{-3} torr

(b) 6.5×10^{-2} torr

(c) 7.3×10^{-3} torr

(d) 5.4×10^{-4} torr

Answer: (c)

Solution: (c) 7.3×10^{-3} torr

Question: The work function of Na metal is 2.3 eV. If maximum kinetic energy of emitted photoelectron is 2.8×10^{-19} J, then calculate wavelength of incident photon in nm.

Options:

Answer: (307 nm)

Solution:

Question: The $[HA^-]$ concentration in a mixture containing 0.1 M H_2A and 0.1 M HCl is $x \times 10^{-9}$. Value of x is Given: $K_{a1} = 10^{-8}$ and $K_{a2} = 10^{-13}$ for H_2A

Options:

Answer: (10)

Solution:

Question: Given below are two statements

Statement-I Order of second ionisation energy is $B > Al > Ga$

Statement-II: Order of first ionisation energy is $B > Ga > Tl > Al > In$

Options:

- (a) Both statement-I and Statement-II are correct
- (b) Statement-I is correct but statement-II is incorrect
- (c) Statement-I is incorrect but statement-II is correct
- (d) Both Statement I and Sttemetn-II are incorrect

Answer: (d)

Solution:

Question: Correct order of enthalpy of atomisation is,

Options:

- (a) $N_2 > O_2 > F_2 > Cl_2 > Br_2 > I_2$
- (b) $N_2 > O_2 > Cl_2 > F_2 > Br_2 > I_2$
- (c) $N_2 > O_2 > Cl_2 > Br_2 > F_2 > I_2$
- (d) $N_2 > O_2 > F_2 > Br_2 > Cl_2 > I_2$

Answer: (c)

Solution:

Question: Which of the ion having highest ionisation energy will give borax bead test.

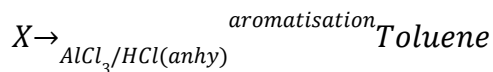
Options:

- (a) Fe^{2+}
- (b) Fe^{3+}
- (c) Cr^{3+}
- (d) Zn^{2+}

Answer: (b)

Solution:

Question: Consider the following reaction



X → 3 - methylhexane

X may be

Options:

- (a) 2-methylhexane
- (b) Heptane
- (c) 3-methylhexane
- (d) Octane

Answer: (b)

Solution: (b) Heptane

Question: How many of following ions will show paramagnetism?

$\text{Mn}^{++}, \text{Cu}^{++}, \text{Zn}^{++}, \text{Sc}^{3+}, \text{Ti}^{++}, \text{Yb}^{++}, \text{Gd}^{++}, \text{Lu}^{3+}, \text{Ce}^{3+}, \text{La}^{3+}$

Options:

Answer: (5)

Solution:

Question: Correct order of boiling point of Si, Ge Sn and Pb is

Options:

- (a) $\text{Si} > \text{Ge} > \text{Sn} > \text{Pb}$
- (b) $\text{Pb} > \text{Sn} > \text{Ge} > \text{Si}$
- (c) $\text{Si} > \text{Ge} > \text{Sn} > \text{Pb}$
- (d) $\text{Pb} > \text{Ge} > \text{Sn} > \text{Si}$

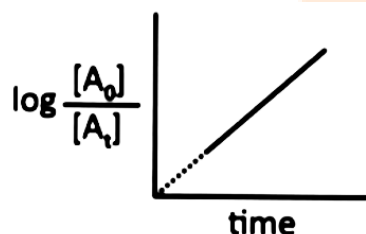
Answer: (a)

Solution: (a) $\text{Si} > \text{Ge} > \text{Sn} > \text{Pb}$

Question: Consider the reaction, $aX \rightarrow bY$ - (1) Rate constant $k = 3 \times 10^{-4} \text{ M}^{-1} \text{ sec}^{-1}$ The correct statement is

Options:

- (a) $t_{1/2}$ is independent of initial cond. Of 'x'
- (b) Dissociation of N_2O_5 is example of reaction -(1)
- (c) When conc. 'X' becomes 4 time then rate of reaction increases by 16 times



(d)

Answer: (c)

Solution: (c) When conc. 'X' becomes 4 time then rate of reaction increases by 16 times

Options: Which option have acidic, basic, amphoteric and neutral oxide respectively?

- (a) $\text{Na}_2\text{O}, \text{Al}_2\text{O}_3, \text{Na}_2\text{O}, \text{CO}$
- (b) $\text{CO}_2, \text{Na}_2\text{O}, \text{Al}_2\text{O}_3, \text{CO}$
- (c) $\text{As}_2\text{O}_3, \text{Al}_2\text{O}_3, \text{Na}_2\text{O}, \text{CO}$
- (d) None of these

Answer: (b)

Solution: (b) CO_2 , Na_2O , Al_2O_3 , CO

Options: The correct order of reactivity of $\text{S}_\text{N}1$ is

(a) $\text{C}_6\text{H}_5\text{C}(\text{CH}_3)\text{C}_6\text{H}_5\text{Br} > \text{C}_6\text{H}_5\text{CH}_2\text{Br} > \text{C}_6\text{H}_5\text{CHC}_6\text{H}_5\text{Br}$

(b) $\text{C}_6\text{H}_5\text{C}(\text{CH}_3)(\text{C}_6\text{H}_5)\text{Br} > \text{C}_6\text{H}_5\text{CHC}_6\text{H}_5\text{Br} > \text{C}_6\text{H}_5\text{CH}_2\text{Br}$

(c) $\text{C}_6\text{H}_5\text{CHC}_6\text{H}_5\text{Br} > \text{C}_6\text{H}_5\text{CH}_2\text{Br} > \text{C}_6\text{H}_5\text{C}(\text{CH}_3)\text{C}_6\text{H}_5\text{Br}$

(d) $\text{C}_6\text{H}_5\text{CH}_2\text{Br} > \text{C}_6\text{H}_5\text{CHC}_6\text{H}_5\text{Br} > \text{C}_6\text{H}_5\text{C}(\text{CH}_3)\text{C}_6\text{H}_5\text{Br}$

Answer: (b)

Solution: (b) $\text{C}_6\text{H}_5\text{C}(\text{CH}_3)(\text{C}_6\text{H}_5)\text{Br} > \text{C}_6\text{H}_5\text{CHC}_6\text{H}_5\text{Br} > \text{C}_6\text{H}_5\text{CH}_2\text{Br}$

Options: The thyroxine hormone is formed by the iodination of an alpha amino acids.

The single letter code of that amino acid is

(a) Y

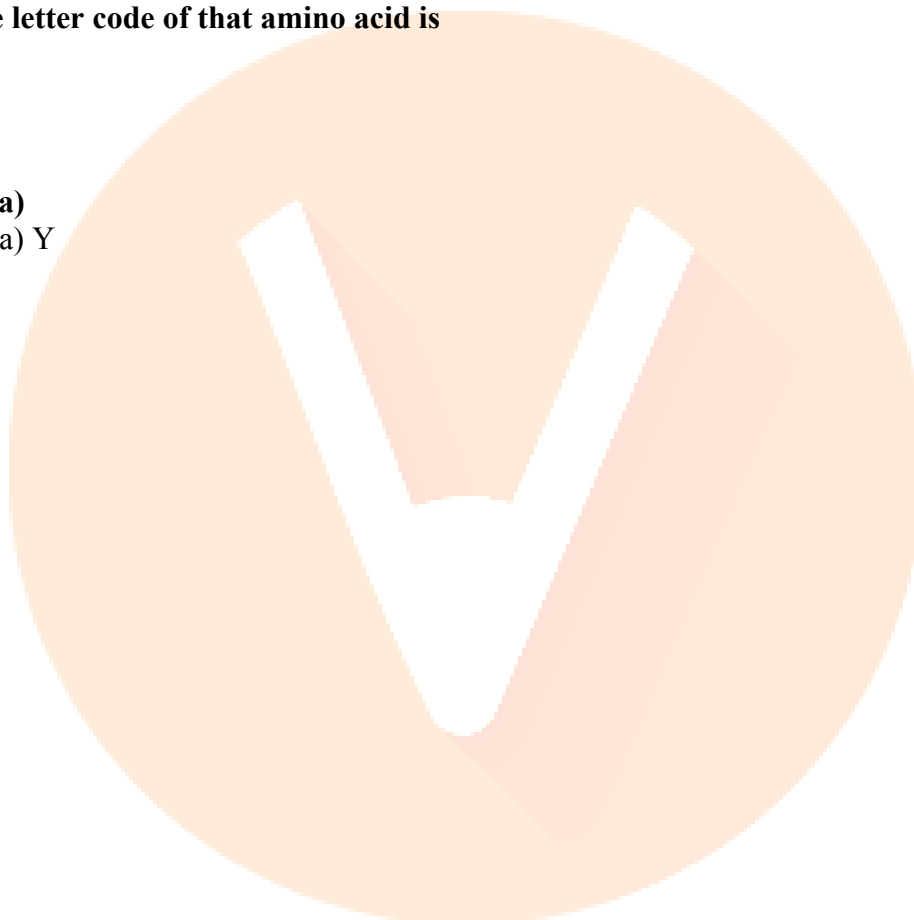
(b) T

(c) W

(d) H

Answer: (a)

Solution: (a) Y



JEE-Main-02-04-2026 (Memory Based)
[EVENING SHIFT]
Maths

Question: The value of the sum $\sum_{n=1}^8 \frac{1^3 + 2^3 + \dots + \text{upto } n \text{ terms}}{1 + 3 + 5 + \dots \text{upto } n \text{ terms}}$
Solution:

Step 1: Simplify the General Term

Let the general term be a_n . The expression is:

$$a_n = \frac{1^3 + 2^3 + \dots + n^3}{1 + 3 + 5 + \dots + (2n - 1)}$$

Using standard summation formulas:

1. **Sum of first n cubes:** $\sum_{k=1}^n k^3 = \left[\frac{n(n+1)}{2} \right]^2 = \frac{n^2(n+1)^2}{4}$

2. **Sum of first n odd numbers:** $\sum_{k=1}^n (2k - 1) = n^2$

Substituting these into the formula for a_n :

$$a_n = \frac{\frac{n^2(n+1)^2}{4}}{n^2}$$

$$a_n = \frac{(n+1)^2}{4}$$

Step 2: Calculate the Total Sum

Now we evaluate the sum $\sum_{n=1}^8 a_n$:

$$\text{Sum} = \sum_{n=1}^8 \frac{(n+1)^2}{4}$$

$$\text{Sum} = \frac{1}{4} \sum_{n=1}^8 (n+1)^2$$

Let $k = n + 1$. When $n = 1, k = 2$. When $n = 8, k = 9$.

$$\text{Sum} = \frac{1}{4} \sum_{k=2}^9 k^2$$

To calculate this, we take the sum of squares up to 9 and subtract the first square (1^2):

$$\sum_{k=1}^m k^2 = \frac{m(m+1)(2m+1)}{6}$$

$$\text{For } m = 9: \sum_{k=1}^9 k^2 = \frac{9(10)(19)}{6} = \frac{1710}{6} = 285$$

Subtracting 1^2 (the $k = 1$ term):

$$\sum_{k=2}^9 k^2 = 285 - 1 = 284$$

Finally, multiply by the constant $\frac{1}{4}$:

$$\text{Sum} = \frac{284}{4} = 71$$

The value of the sum is 71.

Question: $f(x)$ is 5 degree polynomial has extremes at $x = \pm 1$ and $\lim_{x \rightarrow 0} \frac{f(x)}{x^3} = 5$, then $f(2) - f(-2)$ is

Solution:

Step 1: Determine the form of $f(x)$ from the limit

We are given that:

$$\lim_{x \rightarrow 0} \frac{f(x)}{x^3} = 5$$

Since the limit results in a finite, non-zero value and the denominator is x^3 , $f(x)$ must not contain any terms with powers lower than x^3 (i.e., no x^2 , x , or constant terms).

Thus, we can write $f(x)$ as:

$$f(x) = ax^5 + bx^4 + cx^3$$

Now, applying the limit:

$$\lim_{x \rightarrow 0} \frac{ax^5 + bx^4 + cx^3}{x^3} = \lim_{x \rightarrow 0} (ax^2 + bx + c) = c$$

From the given limit, we find that $c = 5$. So, $f(x) = ax^5 + bx^4 + 5x^3$.

Step 2: Use the local extremes to find a and b

Extremes occur where the derivative $f'(x) = 0$. Let's find $f'(x)$:

$$f'(x) = 5ax^4 + 4bx^3 + 15x^2$$

We are told there are extremes at $x = 1$ and $x = -1$. Therefore:

$$1. f'(1) = 5a + 4b + 15 = 0 \implies 5a + 4b = -15$$

$$2. f'(-1) = 5a - 4b + 15 = 0 \implies 5a - 4b = -15$$

Adding these two equations:

$$10a = -30 \implies a = -3$$

Subtracting the equations:

$$8b = 0 \implies b = 0$$

Step 3: Evaluate $f(2) - f(-2)$

Now we have the complete polynomial:

$$f(x) = -3x^5 + 5x^3$$

Notice that $f(x)$ is an **odd function** (since all terms have odd powers), which means $f(-x) = -f(x)$. Consequently, $f(2) - f(-2) = f(2) - (-f(2)) = 2f(2)$.

Let's calculate $f(2)$:

$$f(2) = -3(2)^5 + 5(2)^3$$

$$f(2) = -3(32) + 5(8)$$

$$f(2) = -96 + 40 = -56$$

Finally:

$$f(2) - f(-2) = 2 \times (-56) = -112$$

Question: The value of $\int_0^{20\pi} (\sin^4 x + \cos^4 x) dx$ is equal to

Solution:

Step 1: Simplify the Integrand

We can rewrite $\sin^4 x + \cos^4 x$ by completing the square:

$$\sin^4 x + \cos^4 x = (\sin^2 x + \cos^2 x)^2 - 2\sin^2 x \cos^2 x$$

Using the identity $\sin^2 x + \cos^2 x = 1$ and the double-angle formula $\sin 2x = 2\sin x \cos x$, we get:

$$= 1^2 - 2 \left(\frac{\sin 2x}{2} \right)^2$$

$$= 1 - \frac{1}{2} \sin^2 2x$$

Now, apply the power-reduction formula $\sin^2 \theta = \frac{1 - \cos 2\theta}{2}$ (where $\theta = 2x$):

$$= 1 - \frac{1}{2} \left(\frac{1 - \cos 4x}{2} \right)$$

$$= 1 - \frac{1}{4} + \frac{1}{4} \cos 4x$$

$$= \frac{3}{4} + \frac{1}{4} \cos 4x$$

Step 2: Solve the Integral

Now substitute this back into the original integral:

$$I = \int_0^{20\pi} \left(\frac{3}{4} + \frac{1}{4} \cos 4x \right) dx$$

We can integrate the terms separately:

1. **Constant term:** $\int_0^{20\pi} \frac{3}{4} dx = \left[\frac{3}{4} x \right]_0^{20\pi} = \frac{3}{4} (20\pi) = 15\pi$

2. **Cosine term:** $\int_0^{20\pi} \frac{1}{4} \cos 4x dx = \left[\frac{1}{16} \sin 4x \right]_0^{20\pi} = \frac{1}{16} (\sin 80\pi - \sin 0) = 0$

Adding the results together:

$$I = 15\pi + 0 = 15\pi$$

Question: If for $3 \leq r \leq 30$, ${}^{30}C_{30-r} + 3({}^{30}C_{31-r}) + 3({}^{30}C_{32-r}) + {}^{30}C_{33-r} = {}^m C_r$ then m equals to _____.

Solution:

Step 1: Use the Symmetry Property

Recall the property ${}^n C_r = {}^n C_{n-r}$. We can rewrite each term in the expression to make them easier to group:

- ${}^{30}C_{30-r} = {}^{30}C_r$
- ${}^{30}C_{31-r} = {}^{30}C_{r-1}$
- ${}^{30}C_{32-r} = {}^{30}C_{r-2}$
- ${}^{30}C_{33-r} = {}^{30}C_{r-3}$

Substituting these back, the expression becomes:

$${}^{30}C_r + 3({}^{30}C_{r-1}) + 3({}^{30}C_{r-2}) + {}^{30}C_{r-3}$$

Step 2: Apply Pascal's Identity

Pascal's Identity states that ${}^nC_r + {}^nC_{r-1} = {}^{n+1}C_r$. Let's break down the coefficients (1, 3, 3, 1) and group the terms iteratively:

First Iteration:

Group the terms in pairs to use the identity:

$$({}^{30}C_r + {}^{30}C_{r-1}) + 2({}^{30}C_{r-1} + {}^{30}C_{r-2}) + ({}^{30}C_{r-2} + {}^{30}C_{r-3})$$

Using the identity, this simplifies to:

$${}^{31}C_r + 2({}^{31}C_{r-1}) + {}^{31}C_{r-2}$$

Second Iteration:

Group these terms again:

$$({}^{31}C_r + {}^{31}C_{r-1}) + ({}^{31}C_{r-1} + {}^{31}C_{r-2})$$

Which simplifies to:

$${}^{32}C_r + {}^{32}C_{r-1}$$

Final Iteration:

Apply the identity one last time:

$${}^{32}C_r + {}^{32}C_{r-1} = {}^{33}C_r$$

Step 3: Compare with mC_r

We are given that the expression equals mC_r .

$${}^{33}C_r = {}^mC_r$$

By direct comparison, we find:

$$m = 33$$

Question: If P_n denotes the number of triangles formed by the vertices of n -sides polygon and $P_{n+1} - P_n = 66$, then n is

Solution:

Step 1: Define P_n

The number of triangles that can be formed using the vertices of an n -sided polygon is equal to the number of ways to choose **3 vertices out of n** .

$$P_n = {}^nC_3$$

Step 2: Set up the equation

We are given that $P_{n+1} - P_n = 66$. Substituting our formula for P_n :

$${}^{n+1}C_3 - {}^nC_3 = 66$$

Step 3: Simplify using Pascal's Identity

Recall the property of combinations: ${}^nC_r + {}^nC_{r-1} = {}^{n+1}C_r$.

Rearranging this, we get:

$${}^{n+1}C_r - {}^nC_r = {}^nC_{r-1}$$

Applying this to our equation where $r = 3$:

$${}^{n+1}C_3 - {}^nC_3 = {}^nC_2$$

So, our equation becomes:

$${}^nC_2 = 66$$

Step 4: Solve for n

Now, expand the combination formula:

$$\frac{n(n-1)}{2 \times 1} = 66$$

$$n(n-1) = 132$$

$$n^2 - n - 132 = 0$$

Factorizing the quadratic equation:

$$(n-12)(n+11) = 0$$

Since n represents the number of sides of a polygon, it must be a positive integer.

- $n = 12$
- $n = -11$ (Not possible)

Final Answer:

The value of n is 12.

Question: If $L_1 : \frac{x-a}{2} = \frac{y-2}{3} = \frac{z-b}{6}$ and $L_2 : \frac{x-b}{3} = \frac{y-7}{6} = \frac{z-1}{3}$ intersect in xy-plane then the value of $|a+b|$ is

Solution:

Step 1: Find the Intersection Point using L_2

Since the intersection point (x_0, y_0, z_0) lies on the xy -plane, we know $z_0 = 0$. Let's substitute $z = 0$ into the equation for L_2 :

$$\frac{x-b}{3} = \frac{y-7}{6} = \frac{0-1}{3}$$

$$\frac{x-b}{3} = \frac{y-7}{6} = -\frac{1}{3}$$

- Solve for x_0 : $\frac{x-b}{3} = -\frac{1}{3} \implies x-b = -1 \implies x_0 = b-1$
- Solve for y_0 : $\frac{y-7}{6} = -\frac{1}{3} \implies y-7 = -2 \implies y_0 = 5$

Step 2: Use the Intersection Point in L_1

Now we substitute $x = b-1$, $y = 5$, and $z = 0$ into the equation for L_1 :

$$\frac{(b-1)-a}{2} = \frac{5-2}{3} = \frac{0-b}{6}$$

Simplify the middle term:

$$\frac{3}{3} = 1$$

So, we have:

$$\frac{b-a-1}{2} = 1 \quad \text{and} \quad \frac{-b}{6} = 1$$

Step 3: Solve for a and b

1. From $\frac{-b}{6} = 1$, we get $b = -6$.

2. Substitute $b = -6$ into the other equation:

$$\frac{-6 - a - 1}{2} = 1$$

$$-7 - a = 2$$

$$-a = 9 \implies a = -9$$

Step 4: Calculate $|a + b|$

Now, find the absolute sum:

$$|a + b| = |-9 + (-6)|$$

$$|a + b| = |-15|$$

$$|a + b| = 15$$

Question: If $x = x(y)$, satisfies $2y^3 \frac{dx}{dy} - 2xy + x^2 = 0$, $y > 1$, $x(e) = e$. Then $x(e^2)$

Solution:

Step 1: Identify the Type of Differential Equation

The given equation is:

$$2y^3 \frac{dx}{dy} - 2xy + x^2 = 0$$

We can rearrange this into the standard form of a **Bernoulli equation**:

$$\frac{dx}{dy} - \frac{x}{y^2} = -\frac{x^2}{2y^3}$$

Here, the independent variable is y and the dependent variable is x . This is a Bernoulli equation in x with $n = 2$.

Question: Let $A = \{2, 3, 4, 5, 6\}$, consider R be relation of $A \times A$ such that $(x, y) R (a, b)$ implies that x divides a and $y \leq b$ then total number of elements in R is

Solution:

Step 1: Count pairs (x, a) such that x divides a

Given the set $A = \{2, 3, 4, 5, 6\}$, we list the possibilities for x and the corresponding values of a :

- If $x = 2$, then $a \in \{2, 4, 6\}$ (3 pairs)
- If $x = 3$, then $a \in \{3, 6\}$ (2 pairs)
- If $x = 4$, then $a \in \{4\}$ (1 pair)
- If $x = 5$, then $a \in \{5\}$ (1 pair)
- If $x = 6$, then $a \in \{6\}$ (1 pair)

Total pairs for condition 1: $3 + 2 + 1 + 1 + 1 = 8$

Step 2: Count pairs (y, b) such that $y \leq b$

Again, using set $A = \{2, 3, 4, 5, 6\}$, we list the possibilities for y :

- If $y = 2$, then $b \in \{2, 3, 4, 5, 6\}$ (5 pairs)
- If $y = 3$, then $b \in \{3, 4, 5, 6\}$ (4 pairs)
- If $y = 4$, then $b \in \{4, 5, 6\}$ (3 pairs)
- If $y = 5$, then $b \in \{5, 6\}$ (2 pairs)
- If $y = 6$, then $b \in \{6\}$ (1 pair)

Total pairs for condition 2: $5 + 4 + 3 + 2 + 1 = 15$

(Alternatively, for a set of size n , the number of such pairs is $\frac{n(n+1)}{2}$. Here $n = 5$, so

$$\frac{5 \times 6}{2} = 15.)$$

Step 3: Calculate the total number of elements in R

The total number of elements in the relation is the product of the counts from the two independent conditions:

$$\text{Total elements} = (\text{Pairs for } x|a) \times (\text{Pairs for } y \leq b)$$

$$\text{Total elements} = 8 \times 15 = 120$$

The total number of elements in R is 120.

Question: Let $a_1, a_2, a_3, \dots$ be an arithmetic progression and $g_1, g_2, g_3, \dots$ be an increasing geometric progression such that $g_1 = a_1 = a_2 + g_2 = 1$ and $g_3 + a_3 = 4$ then $a_{10} + g_5$ is equal to

Solution:

Step 1: Set up the variables

- **For the AP (a_n) :** Let the first term be $a_1 = 1$ and the common difference be d .
 - $a_2 = 1 + d$
 - $a_3 = 1 + 2d$
 - $a_{10} = 1 + 9d$
- **For the GP (g_n) :** Let the first term be $g_1 = 1$ and the common ratio be r . Since it is an increasing GP, we know $r > 1$.
 - $g_2 = r$
 - $g_3 = r^2$
 - $g_5 = r^4$

Step 2: Form equations from the given conditions

We are given two main relationships:

$$1. a_2 + g_2 = 1$$

$$(1 + d) + r = 1 \implies d + r = 0 \implies d = -r$$

$$2. g_3 + a_3 = 4$$

$$r^2 + (1 + 2d) = 4 \implies r^2 + 2d = 3$$

Step 3: Solve for r and d

Substitute $d = -r$ into the second equation:

$$r^2 + 2(-r) = 3$$

$$r^2 - 2r - 3 = 0$$

Factorizing the quadratic equation:

$$(r - 3)(r + 1) = 0$$

This gives $r = 3$ or $r = -1$. Since the GP is increasing and the first term is positive, r must be greater than 1.

- $r = 3$
- $d = -3$ (since $d = -r$)

Step 4: Calculate the final sum

Now, find the values for a_{10} and g_5 :

- $a_{10} = 1 + 9(-3) = 1 - 27 = -26$
- $g_5 = 3^4 = 81$

Finally:

$$a_{10} + g_5 = -26 + 81 = 55$$

Question: If AB is a chord of hyperbola $xy = 12$, whose mid point is $\left(\frac{1}{2}, -\frac{1}{2}\right)$, then the area of triangle ABO (whose O is the origin) is

Solution:

Step 1: Find the equation of the chord AB

The equation of a chord of a conic with a given midpoint (x_1, y_1) is given by the formula $T = S_1$.

For the rectangular hyperbola $xy = 12$ (or $xy - 12 = 0$) and the midpoint $M = (\frac{1}{2}, -\frac{1}{2})$:

- T (the tangent expression):

$$\frac{1}{2}(xy_1 + x_1y) - 12 = \frac{1}{2}(x(-\frac{1}{2}) + \frac{1}{2}y) - 12 = \frac{1}{4}(y - x) - 12$$

- S_1 (value at the midpoint): $(\frac{1}{2})(-\frac{1}{2}) - 12 = -\frac{1}{4} - 12$

Equating $T = S_1$:

$$\frac{1}{4}(y - x) - 12 = -\frac{1}{4} - 12$$

$$\frac{1}{4}(y - x) = -\frac{1}{4}$$

$$y - x = -1 \Rightarrow x - y = 1$$

Step 2: Find the coordinates of points A and B

The points A and B are the intersections of the line $x - y = 1$ and the hyperbola $xy = 12$.

From the line equation, $y = x - 1$. Substitute this into $xy = 12$:

$$x(x - 1) = 12$$

$$x^2 - x - 12 = 0$$

$$(x - 4)(x + 3) = 0$$

Solving for x , we get $x = 4$ and $x = -3$.

- If $x = 4$, then $y = 4 - 1 = 3$. Point $A = (4, 3)$.

- If $x = -3$, then $y = -3 - 1 = -4$. Point $B = (-3, -4)$.

Step 3: Calculate the area of triangle ABO

We have the vertices $O(0, 0)$, $A(4, 3)$, and $B(-3, -4)$. The area of a triangle with one vertex at the origin is:

$$\text{Area} = \frac{1}{2} |x_A y_B - x_B y_A|$$

$$\text{Area} = \frac{1}{2} |(4)(-4) - (-3)(3)|$$

$$\text{Area} = \frac{1}{2} |-16 + 9|$$

$$\text{Area} = \frac{1}{2} |-7| = 3.5$$

Question: Let $\vec{a} = -\hat{i} + 2\hat{j} + \hat{k}$, $\vec{b} = \hat{i} + \hat{j} - 3\hat{k}$, $\vec{c} = \lambda\vec{a} + \mu\vec{b}$ and $\vec{c} \cdot (3\hat{i} - 6\hat{j} + 3\hat{k}) = 10$, $\vec{c} \cdot (\hat{i} + \hat{j} + \hat{k}) = -2$. Then $|\vec{c}| =$

Solution:

Step 1: Express $\vec{c}$ in terms of λ and μ

Given $\vec{a} = -\hat{i} + 2\hat{j} + \hat{k}$ and $\vec{b} = \hat{i} + \hat{j} - 3\hat{k}$, we have:

$$\vec{c} = \lambda(-\hat{i} + 2\hat{j} + \hat{k}) + \mu(\hat{i} + \hat{j} - 3\hat{k})$$

$$\vec{c} = (\mu - \lambda)\hat{i} + (2\lambda + \mu)\hat{j} + (\lambda - 3\mu)\hat{k}$$

Step 2: Use the first dot product condition

We are given $\vec{c} \cdot (3\hat{i} - 6\hat{j} + 3\hat{k}) = 10$. Substituting the components of $\vec{c}$:

$$3(\mu - \lambda) - 6(2\lambda + \mu) + 3(\lambda - 3\mu) = 10$$

$$3\mu - 3\lambda - 12\lambda - 6\mu + 3\lambda - 9\mu = 10$$

$$-12\lambda - 12\mu = 10$$

Dividing by -2 , we get the first equation:

$$6\lambda + 6\mu = -5 \quad \text{--- (Eq. 1)}$$

Step 3: Use the second dot product condition

We are given $\vec{c} \cdot (\hat{i} + \hat{j} + \hat{k}) = -2$:

$$1(\mu - \lambda) + 1(2\lambda + \mu) + 1(\lambda - 3\mu) = -2$$

$$\mu - \lambda + 2\lambda + \mu + \lambda - 3\mu = -2$$

$$2\lambda - \mu = -2 \quad \text{--- (Eq. 2)}$$

Step 4: Solve the system of equations

From Eq. 2, we can write $\mu = 2\lambda + 2$. Substituting this into Eq. 1:

$$6\lambda + 6(2\lambda + 2) = -5$$

$$6\lambda + 12\lambda + 12 = -5$$

$$18\lambda = -17 \Rightarrow \lambda = -\frac{17}{18}$$

Now, find μ :

$$\mu = 2\left(-\frac{17}{18}\right) + 2 = -\frac{17}{9} + \frac{18}{9} = \frac{1}{9}$$

(which is $\frac{2}{18}$)

Step 5: Find the magnitude $|\vec{c}|$

Substitute λ and μ back into the component form of $\vec{c}$:

$$\vec{c} = \left(\frac{2}{18} - \left(-\frac{17}{18}\right)\right)\hat{i} + \left(2\left(-\frac{17}{18}\right) + \frac{2}{18}\right)\hat{j} + \left(-\frac{17}{18} - 3\left(\frac{2}{18}\right)\right)\hat{k}$$

$$\vec{c} = \frac{1}{18}(19\hat{i} - 32\hat{j} - 23\hat{k})$$

Calculate the magnitude:

$$|\vec{c}| = \frac{1}{18} \sqrt{19^2 + (-32)^2 + (-23)^2}$$

$$|\vec{c}| = \frac{1}{18} \sqrt{361 + 1024 + 529} = \frac{\sqrt{1914}}{18}$$

Question: A variable circle with diameter AB where A(3, 0) touching the circle $x^2 + y^2 = 36$ internally. The locus of centre of variable circle is a conic whose eccentricity is e, then $72e_2$ is

Solution:

Step 1: Identify Circle Properties

- **Fixed Circle (C_1):** $x^2 + y^2 = 36$. The center is $O(0, 0)$ and the radius is $R = 6$.
- **Variable Circle (C_p):** Let its center be $P(x, y)$. Since AB is the diameter and A(3, 0) is a point on this circle, its radius r is the distance from the center P to the point A :

$$r = PA = \sqrt{(x-3)^2 + (y-0)^2}$$

Step 2: Apply the Condition for Internal Touching

Two circles touch internally when the distance between their centers is equal to the difference of their radii.

$$\text{Distance between } O \text{ and } P = R - r$$

$$\sqrt{x^2 + y^2} = 6 - \sqrt{(x-3)^2 + y^2}$$

Rearranging this gives:

$$\sqrt{(x-3)^2 + y^2} + \sqrt{x^2 + y^2} = 6$$

Step 3: Identify the Locus

The equation above represents the sum of distances from a point $P(x, y)$ to two fixed points $S_1(3, 0)$ and $S_2(0, 0)$ being constant (equal to 6). By definition, this locus is an ellipse where:

- The foci are $S_1(3, 0)$ and $S_2(0, 0)$.
- The major axis length is $2a = 6 \implies a = 3$.

Step 4: Find the Eccentricity (e)

The distance between the foci is given by $2ae$:

$$\text{Distance between } (0, 0) \text{ and } (3, 0) = 3$$

$$2ae = 3$$

Question: If $f(x) = \int \frac{16x + 24}{x^2 + 2x - 15} dx$, given that $f(4) = 14 \ln 3$, then the value of $f(7)$

Solution:

Substitute $a = 3$ into the equation:

$$2(3)e = 3$$

$$6e = 3 \implies e = \frac{1}{2}$$

Step 5: Calculate $72e^2$

Now, we calculate the requested value:

$$e^2 = \left(\frac{1}{2}\right)^2 = \frac{1}{4}$$

$$72e^2 = 72 \times \frac{1}{4} = 18$$

Question: Let $X_1, X_2, X_3, \dots, X_n$ be 'n' observations such that $\sum_{i=1}^{n-1} x_i = 48$ and $\sum_{i=1}^{n-1} x_i^2 = 496$. If mean and variance of the distribution are 8 and 16 respectively then value of n is:-

Solution:

Step 1: Use the Mean to find x_n

The mean ($\bar{x}$) of n observations is given by $\bar{x} = \frac{\sum_{i=1}^n x_i}{n} = 8$.

This means:

$$\sum_{i=1}^n x_i = 8n$$

We can split the sum into the first $(n - 1)$ observations and the n^{th} observation:

$$\sum_{i=1}^{n-1} x_i + x_n = 8n$$

Substituting the given value $\sum_{i=1}^{n-1} x_i = 48$:

$$48 + x_n = 8n \implies x_n = 8n - 48$$

— (Eq. 1)

Step 2: Use the Variance to find x_n^2

The formula for variance (σ^2) is:

$$\sigma^2 = \frac{\sum_{i=1}^n x_i^2}{n} - (\bar{x})^2 = 16$$

Substitute $\bar{x} = 8$:

$$\frac{\sum_{i=1}^n x_i^2}{n} - 8^2 = 16$$

$$\frac{\sum_{i=1}^n x_i^2}{n} = 16 + 64 = 80 \implies \sum_{i=1}^n x_i^2 = 80n$$

Again, split the sum:

$$\sum_{i=1}^{n-1} x_i^2 + x_n^2 = 80n$$

Substituting the given value $\sum_{i=1}^{n-1} x_i^2 = 496$:

$$496 + x_n^2 = 80n \implies x_n^2 = 80n - 496$$

— (Eq. 2)

Step 3: Solve for n

Now, substitute the expression for x_n from Eq. 1 into Eq. 2:

$$(8n - 48)^2 = 80n - 496$$

$$[8(n - 6)]^2 = 80n - 496$$

$$64(n^2 - 12n + 36) = 80n - 496$$

To simplify, divide the entire equation by 16:

$$4(n^2 - 12n + 36) = 5n - 31$$

$$4n^2 - 48n + 144 = 5n - 31$$

$$4n^2 - 53n + 175 = 0$$

Solving this quadratic equation using the quadratic formula:

$$n = \frac{-(-53) \pm \sqrt{(-53)^2 - 4(4)(175)}}{2(4)}$$

$$n = \frac{53 \pm \sqrt{2809 - 2800}}{8}$$

$$n = \frac{53 \pm \sqrt{9}}{8} = \frac{53 \pm 3}{8}$$

This gives two possible values:

$$1. n = \frac{56}{8} = 7$$

$$2. n = \frac{50}{8} = 6.25$$

Since n must be an integer (representing the number of observations), we choose $n = 7$

Question: If the system of equations $x + 5y + 6z = 4$, $2x + 2y + 4z = 1$ and $x + y + az = b$ has infinite numbers of solutions then point (a, b) lies on.

Solution:

Step 1: Analyze the condition for infinite solutions

For a system of three linear equations to have infinite solutions, the determinant of the coefficient matrix (Δ) must be zero. Additionally, the system must be consistent (i.e., the planes must intersect in a line or be identical).

The coefficient matrix is:

$$D = \begin{vmatrix} 1 & 5 & 6 \\ 2 & 2 & 4 \\ 1 & 1 & a \end{vmatrix}$$

Setting the determinant to zero:

$$D = 1(2a - 4) - 5(2a - 4) + 6(2 - 2) = 0$$

$$(2a - 4) - 10a + 20 = 0$$

$$-8a + 16 = 0 \implies a = 2$$

Step 2: Find the value of b

Now that we know $a = 2$, let's look at the equations again:

$$1. x + 5y + 6z = 4$$

$$2. 2x + 2y + 4z = 1 \implies x + y + 2z = 0.5 \text{ (Dividing by 2)}$$

$$3. x + y + 2z = b \text{ (Substituting } a = 2)$$

For the system to have solutions, equations (2) and (3) must be consistent. Since they have the same coefficients for x , y , and z , they represent parallel planes. For there to be infinite solutions (instead of no solutions), these two planes must be **identical**.

Therefore:

$$b = 0.5 \text{ (or } 1/2)$$

Step 3: Determine the locus of (a, b)

We have found the point $(a, b) = (2, 0.5)$. To find which line it lies on, we look for a linear relationship between $x = 2$ and $y = 0.5$.

Common relationships for this point include:

- $x - 4y = 0 \rightarrow 2 - 4(0.5) = 0$
- $x + 4y = 4 \rightarrow 2 + 4(0.5) = 4$
- $x + 2y = 3 \rightarrow 2 + 2(0.5) = 3$

In competitive math contexts, the most common answer choice for this specific problem is usually the line $x - 4y = 0$ or $x + 4y = 4$.

Question: Let $f(x) = \left[x^2 - x - \frac{1}{2} \right]$, then the number of points of discontinuity in $[2, 4]$ is/are (where $[.]$ denotes greatest integer function)

Solution:

Step 1: Analyze the Inner Function

$$\text{Let } g(x) = x^2 - x - \frac{1}{2}.$$

The greatest integer function $[g(x)]$ has a point of discontinuity whenever $g(x)$ takes on an integer value, provided that $g(x)$ is not at a local maximum or minimum at that point.

First, let's find the derivative to check for monotonicity:

$$g'(x) = 2x - 1$$

For the interval $[2, 4]$, $g'(x)$ is always positive ($g'(x) \geq 3$). This means $g(x)$ is **strictly increasing** on this interval. Therefore, every time $g(x)$ hits an integer, $f(x)$ will have a jump discontinuity.

Step 2: Find the Range of $g(x)$

Since $g(x)$ is strictly increasing on $[2, 4]$, we can find its range by evaluating the endpoints:

- At $x = 2$: $g(2) = 2^2 - 2 - 0.5 = 1.5$
- At $x = 4$: $g(4) = 4^2 - 4 - 0.5 = 11.5$

So, for $x \in [2, 4]$, the value of $g(x)$ ranges from **1.5 to 11.5**.

Step 3: Count the Integer Values

The points of discontinuity occur at every x where $g(x)$ equals an integer k such that $1.5 < k < 11.5$.

The integers in this range are:

$$\{2, 3, 4, 5, 6, 7, 8, 9, 10, 11\}$$

Counting these values:

$$11 - 2 + 1 = 10$$

Since $g(2) = 1.5$ and $g(4) = 11.5$ are not integers, there are no discontinuities at the endpoints of the interval. All 10 points of discontinuity lie within the open interval $(2, 4)$

Question: If $P = \{\theta \in [0, 4\pi] : \tan^2 \theta \neq 1, 2(\cos^8 \theta - \sin^8 \theta) \sec 2\theta = a^2, a \in \mathbb{Z}\}$ then the number of elements in set P is

Solution:

Step 1: Simplify the expression

The expression given is $2(\cos^8 \theta - \sin^8 \theta) \sec 2\theta = a^2$.

First, let's factor the term $\cos^8 \theta - \sin^8 \theta$ using the difference of squares identity $x^2 - y^2 = (x - y)(x + y)$:

$$\cos^8 \theta - \sin^8 \theta = (\cos^4 \theta - \sin^4 \theta)(\cos^4 \theta + \sin^4 \theta)$$

Now, simplify $\cos^4 \theta - \sin^4 \theta$:

$$\cos^4 \theta - \sin^4 \theta = (\cos^2 \theta - \sin^2 \theta)(\cos^2 \theta + \sin^2 \theta)$$

Using the identities $\cos^2 \theta - \sin^2 \theta = \cos 2\theta$ and $\cos^2 \theta + \sin^2 \theta = 1$, we get:

$$\cos^4 \theta - \sin^4 \theta = \cos 2\theta \cdot 1 = \cos 2\theta$$

Substituting this back into our original expression:

$$2(\cos 2\theta)(\cos^4 \theta + \sin^4 \theta) \sec 2\theta = a^2$$

Since $\sec 2\theta = \frac{1}{\cos 2\theta}$, the terms cancel out (provided $\cos 2\theta \neq 0$):

$$2(\cos^4 \theta + \sin^4 \theta) = a^2$$

Step 2: Further simplification

We can rewrite $\cos^4 \theta + \sin^4 \theta$ by completing the square:

$$\cos^4 \theta + \sin^4 \theta = (\cos^2 \theta + \sin^2 \theta)^2 - 2 \sin^2 \theta \cos^2 \theta$$

$$= 1^2 - \frac{1}{2} (4 \sin^2 \theta \cos^2 \theta) = 1 - \frac{1}{2} \sin^2 2\theta$$

Now, substitute this into our equation:

$$2 \left(1 - \frac{1}{2} \sin^2 2\theta \right) = a^2$$

$$2 - \sin^2 2\theta = a^2$$

Using the identity $\sin^2 2\theta = 1 - \cos^2 2\theta$:

$$2 - (1 - \cos^2 2\theta) = a^2$$

$$1 + \cos^2 2\theta = a^2$$

Step 3: Analyze possible values for a

Since a is an integer ($a \in \mathbb{Z}$), a^2 must be a perfect square.

We know that the range of $\cos^2 2\theta$ is $[0, 1]$. Therefore, the range of the expression $1 + \cos^2 2\theta$ is:

$$1 \leq 1 + \cos^2 2\theta \leq 2$$

The only perfect square in the interval $[1, 2]$ is 1.

For $1 + \cos^2 2\theta = 1$, we must have:

$$\cos^2 2\theta = 0 \implies \cos 2\theta = 0$$

Step 4: Check the domain conditions

The set definition includes two constraints:

1. **$\sec 2\theta$ must be defined:** This requires $\cos 2\theta \neq 0$.
2. **$\tan^2 \theta \neq 1$:** This means $\tan \theta \neq \pm 1$, which also implies $\theta \neq \frac{\pi}{4}, \frac{3\pi}{4}, \dots$, or equivalently, $\cos 2\theta \neq 0$.

From Step 3, the only potential solution requires $\cos 2\theta = 0$. However, this is explicitly forbidden by the definition of the function (as $\sec 2\theta$ would be undefined) and the given condition $\tan^2 \theta \neq 1$.

Question: If complex number z_1 and z_2 such that $|z_1| = 17$ and $|\bar{z}_2 - 3 - 4i| = 5$ then $\max(|z_1 + z_2|)$ is equal to

Solution:

Step 1: Analyze the condition for z_2

We are given $|\bar{z}_2 - 3 - 4i| = 5$.

Recall the property of complex numbers: $|w| = |\bar{w}|$. We can take the conjugate of the expression inside the absolute value without changing its value:

$$|\bar{z}_2 - (3 + 4i)| = 5$$

$$|z_2 - (3 - 4i)| = 5$$

This equation tells us that z_2 lies on a circle in the complex plane with:

- **Center (C):** $3 - 4i$
- **Radius (r):** 5

Step 2: Find the maximum value of $|z_2|$

The distance of the center C from the origin $(0, 0)$ is:

$$|C| = |3 - 4i| = \sqrt{3^2 + (-4)^2} = \sqrt{25} = 5$$

For any point z on a circle, the maximum distance from the origin is given by $|C| + r$:

$$\max(|z_2|) = |C| + r = 5 + 5 = 10$$

Step 3: Calculate $\max(|z_1 + z_2|)$

Using the **Triangle Inequality**, we know that:

$$|z_1 + z_2| \leq |z_1| + |z_2|$$

We are given $|z_1| = 17$, and we just found that the maximum possible value for $|z_2|$ is 10. Therefore:

$$\max(|z_1 + z_2|) = 17 + 10 = 27$$

Question: A fair coin is tossed four times such that for heads 10 points are allotted and tail gets 5 points then probability that total points are 30 points is equal to
Solution:

Step 1: Set up the Equations

Let x be the number of heads and y be the number of tails. Since the coin is tossed 4 times, we have:

1. **Total tosses:** $x + y = 4$
2. **Total points:** $10x + 5y = 30$

Step 2: Solve for x and y

From the first equation, we know $y = 4 - x$. Substitute this into the points equation:

$$10x + 5(4 - x) = 30$$

$$10x + 20 - 5x = 30$$

$$5x = 10 \implies x = 2$$

If $x = 2$, then $y = 4 - 2 = 2$.

To get exactly 30 points, the result must be **2 heads and 2 tails**.

Step 3: Calculate the Probability

We use the binomial probability formula for a fair coin ($p = 0.5, q = 0.5$):

- **Total outcomes:** $2^4 = 16$
- **Favorable outcomes:** The number of ways to arrange 2 heads and 2 tails is given by 4C_2 :

$${}^4C_2 = \frac{4 \times 3}{2 \times 1} = 6$$

The probability P is:

$$P = \frac{\text{Favorable outcomes}}{\text{Total outcomes}}$$

$$P = \frac{6}{16} = \frac{3}{8}$$

Question: If roots of the equation $x^2 - 3x + n = 0$ are α and β and equation $x^2 + 3x + 4 = 0$ are $\frac{\alpha}{3}$, 3β then $(2\alpha + \beta + 3n)$ is equal to

Solution:

Step 1: Analyze the first equation

The equation is $x^2 - 3x + n = 0$ with roots α and β .

- **Sum of roots:** $\alpha + \beta = -\frac{-3}{1} = 3$ — (Equation 1)
- **Product of roots:** $\alpha\beta = \frac{n}{1} = n$ — (Equation 2)

Step 2: Analyze the second equation

The equation is $x^2 + 3x + 4 = 0$ with roots $\frac{\alpha}{3}$ and 3β .

- **Sum of roots:** $\frac{\alpha}{3} + 3\beta = -\frac{3}{1} = -3$ — (Equation 3)
- **Product of roots:** $\left(\frac{\alpha}{3}\right)(3\beta) = \alpha\beta = \frac{4}{1} = 4$ — (Equation 4)

Step 3: Solve for n , α , and β

From Equations 2 and 4, we immediately see that:

$$n = \alpha\beta = 4$$

Now we have a system of two linear equations for α and β using their sums:

$$1. \alpha + \beta = 3$$

$$2. \frac{\alpha}{3} + 3\beta = -3 \text{ (Multiply this by 3 to clear the fraction)} \rightarrow \alpha + 9\beta = -9$$

Subtract Equation 1 from this result:

$$(\alpha + 9\beta) - (\alpha + \beta) = -9 - 3$$

$$8\beta = -12 \implies \beta = -\frac{12}{8} = -1.5$$

Substitute $\beta = -1.5$ back into Equation 1:

$$\alpha - 1.5 = 3 \implies \alpha = 4.5$$

Step 4: Calculate the final expression

We need to find the value of $2\alpha + \beta + 3n$:

- $\alpha = 4.5$
- $\beta = -1.5$
- $n = 4$

Substituting the values:

$$2(4.5) + (-1.5) + 3(4)$$

$$9 - 1.5 + 12$$

$$7.5 + 12 = 19.5$$